

2023



Immutable

Managing 6x Growth with Event-Driven Architecture Using Amazon EventBridge with Immutable

Learn how Immutable in the gaming industry uses event-driven microservices for scalability and reliability using Amazon EventBridge.

[Overview](#) | [Opportunity](#) | [Solution](#) | [Outcome](#) | [AWS Services Used](#)

1,500%

increase in monthly customer transactions in 1 year



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6x

growth in number of employees in first year after transition to event-driven architecture



2

Reduced

undifferentiated heavy lifting

Increased

reliability for better customer experience

Overview

[Immutable](#), a pioneer in Web3 gaming, uses blockchain technology so millions of players can own the digital assets they earn in games. With its unique technology, the unicorn company started growing rapidly, collaborating with industry leaders like GameStop, and incorporating its solutions into games from traditional studios. To accommodate its growth, Immutable transitioned its monolithic architecture, built primarily on Amazon Web Services (AWS), to event-driven microservices that could scale as the company evolved.

Using services such as [Amazon EventBridge](#)—a serverless technology for building event-driven applications at scale across AWS, existing systems, or software-as-a-service applications—Immutable increased scalability to handle a six-times increase in number of employees, improved reliability to enhance the customer experience, and off-loaded the heavy lifting to focus on innovation. By exposing critical events from its solutions in near real time, Immutable built a suite of APIs and developer tools that drive innovation through collaborations across the industry.



Opportunity | Using Amazon EventBridge to Build Solutions Faster and Accelerate Growth for Immutable

Based in Australia, Immutable offers a global game development solution for quickly building and scaling Web3 games. Its suite of turnkey solutions helps game developers get to the marketplace faster with security or player experience. Immutable also develops and publishes its own Web3 games, such as [Guild of Guardians](#).



From its founding in 2018, Immutable built its infrastructure on AWS, leaning heavily on [Amazon Elastic Kubernetes Service](#) (Amazon EKS), a managed service to start, run, and scale Kubernetes. For its databases, the company uses [Amazon Relational Database Service](#) (Amazon RDS), a collection of managed services that organizations can use to set up, operate, and scale databases in the cloud. Although the infrastructure met its needs early on, Immutable grew in 2021 from 50 employees to nearly 300 in less than 1 year. To manage this increasing scale, Immutable's growing engineering teams needed to work on software simultaneously and operate more independently. After evaluating alternatives, Immutable chose to transition to an event-driven architecture using Amazon EventBridge. Because of the compatibility of AWS services, Immutable could build on top of its existing architecture, including adopting [AWS Lambda](#), a serverless, event-driven compute service for running code without thinking about servers or clusters.

Immutable received support from AWS programs and representatives throughout the transition process, including from [AWS Activate](#), a program that provides credits, resources, and expert support to accelerate startups at every stage. In 2022, Immutable completed the transition to an event-driven architecture within 6 months.

Solution | Facilitating Innovation and Supporting a 1,500% Increase in Monthly Customer Transactions in 1 Year Using AWS Serverless Architecture

Using serverless architecture built on Amazon EventBridge and AWS Lambda, Immutable can scale effectively to support its growing team size. More engineering teams can work simultaneously and autonomously on the company's software, increasing the velocity of feature releases and innovation in an emerging Web3 industry. Immutable also benefits from reducing undifferentiated heavy lifting. By building patterns that other teams can adopt, the company's engineering teams are saving time and reducing complexity. Now, they can focus on innovation. "We're trying to build more reliable and scalable systems without increasing the cognitive load on our engineers," says Craig Baker, vice president of architecture at Immutable. "Using Amazon EventBridge, we can build on top of the technology without having to solve complicated problems that aren't core to our business."

Immutable has also seen significant customer growth. From November 2021 to November 2022, the number of monthly customer transactions increased by 1,500 percent. In the same time frame, transactions per collection of



assets increased by 600 percent and average spend per user increased by 50 percent. Immutable's customers have benefited from a more stable, secure, and reliable solution because of the event-driven architecture. Using microservices, Immutable can isolate issues so that only one feature is affected if an error occurs, reducing the impact of any issues and creating a more reliable infrastructure for customers.

Immutable is better poised to launch collaborations with other companies in the industry because its solution can support 9,000 transactions per second with very low latency using AWS serverless technology. For example, in 2022, Illuvium launched its pioneering interoperable blockchain game using Immutable's solution so players can trade nonfungible tokens in less than 1 second without paying gas fees, which players typically pay to validate and process a transaction. "By evolving our architecture on AWS, we saw a 90 percent increase in the trading volume for the Illuvium launch," says Tim Curtin, manager of the site reliability engineering and core engineering teams at Immutable.

Outcome | Expanding to Become a One-Stop Shop for Game Developers Using AWS Services

Immutable's mission is to make asset ownership and commerce in digital gaming worlds real. To fulfill its mission, Immutable is planning multiple product launches and industry partnerships in 2023 and 2024. Immutable and AWS are also working on go-to-market programs to accelerate the onboarding of game studios to Web3 and drive digital ownership to millions of gamers globally.

Immutable aims to be a one-stop shop where Web3 game developers can find virtually all the game components that they need, such as a marketplace and wallet. Immutable plans to continue investing in its infrastructure built on AWS so that the company can scale as needed to support its upcoming offerings such as [Immutable zkEVM](#), a collaboration with Polygon Labs that provides blockchain for games that is low cost, scalable, secure, and compatible with the Ethereum virtual machine, a large service for Web3 developers and decentralized applications. With Ethereum compatibility, Immutable zkEVM aims to expand the customer base, helping more developers build Web3 games without needing to learn a new service. "A microservice architecture centers on scaling engineering teams and the amount of work they can get done," says Baker. "By migrating to an event-driven architecture using Amazon EventBridge, our teams operate independently, deploy independently, and ship software quickly and efficiently."

About Immutable

Australian gaming company Immutable offers a global game development solution for quickly building and scaling Web3 games. Its suite of turnkey solutions helps game developers get to the marketplace faster without diminishing security or player experience.

AWS Services Used



Amazon EventBridge

AWS Step Functions is a visual workflow service that helps developers use AWS services to build distributed applications, automate processes, orchestrate microservices, and create data and machine learning (ML) pipelines.

[Learn more »](#)

AWS Lambda

AWS Lambda is a serverless, event-driven compute service that lets you run code for virtually any type of application or backend service without provisioning or managing servers.

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2024

1 2 3 4 5 6 ... 91

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2

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2

